

Progress Report: AI-based Leak Detection and Localization System

Over the last two weeks, significant progress has been made in model experimentation, data preprocessing, and preparation for simulation/testbed development. The aim remains to design a high-performance, real-time AI solution capable of multi-class leak detection and precise leak localization, backed by both a digital testbed and an intuitive GUI.

Problem Definition (Confirmed in Phase 1)

1. **Multi-class Classification):** Identify leak type from multi-sensor data with classes: No Leak, CC, GL, LC, OL.
2. **Regression:** Estimate leak location with target error ≤ 0.5 m.
3. **Sensors Used:** Accelerometer, Hydrophone, Dynamic Pressure Transducer.
4. **Performance Targets:** $\geq 90\%$ detection accuracy (train & validation), low localization error, low latency (real-time feasibility).

Data Preprocessing and Feature Extraction

1. **Scaling and Normalization** applied to raw time-series data to ensure uniform range and stability in training.
2. **Continuous Wavelet Transform (CWT) Images** generated from raw sensor data for time-frequency analysis. Applied log normalization to enhance contrast and normalize feature distribution.
3. **Data Splitting Strategy:** Stratified split (80% train / 20% validation) to ensure class balance.
4. **Observation:** Pressure + Accelerometer combination yields better classification performance compared to the inclusion of Hydrophone data (which initially reduced accuracy).

Model Development Progress

Baseline Hybrid Model

Architecture: 1D CNN → LSTM → Dense → Output (Softmax for classification). Achieved ~88% validation accuracy in initial tests (Pressure + Accelerometer data). Adding Hydrophone data dropped accuracy to ~62–63%, even after deeper layers and attention mechanisms. Extensive tuning (~30+ training runs) was attempted without satisfactory improvement.

Advanced Model Experiments

- **CNN + LSTM + Attention Layer:** Helped highlight important temporal features, but not enough to recover the Hydrophone-related drop in performance.
- **RNN-LSTM Variant:** Under consideration for better handling of sequential dependencies.

Transfer Learning Approach (NEW)

Used pre-trained CNN backbones (trained on large-scale datasets) and fine-tuned the last layers on our CWT dataset. An initial result of 84% accuracy in just 5 epochs indicates strong potential for reaching $\geq 90\%$ accuracy with full tuning. Plan to integrate a custom attention layer into a pre-trained architecture to satisfy engineering novelty requirements.

Future Model Plans

1. Evaluate Vision Transformers (ViT) for state-of-the-art time–frequency image classification.
2. Explore a hybrid physics + data-driven model (Domain Adaptation) as per supervisor feedback.

Simulation/Testbed & GUI Plans

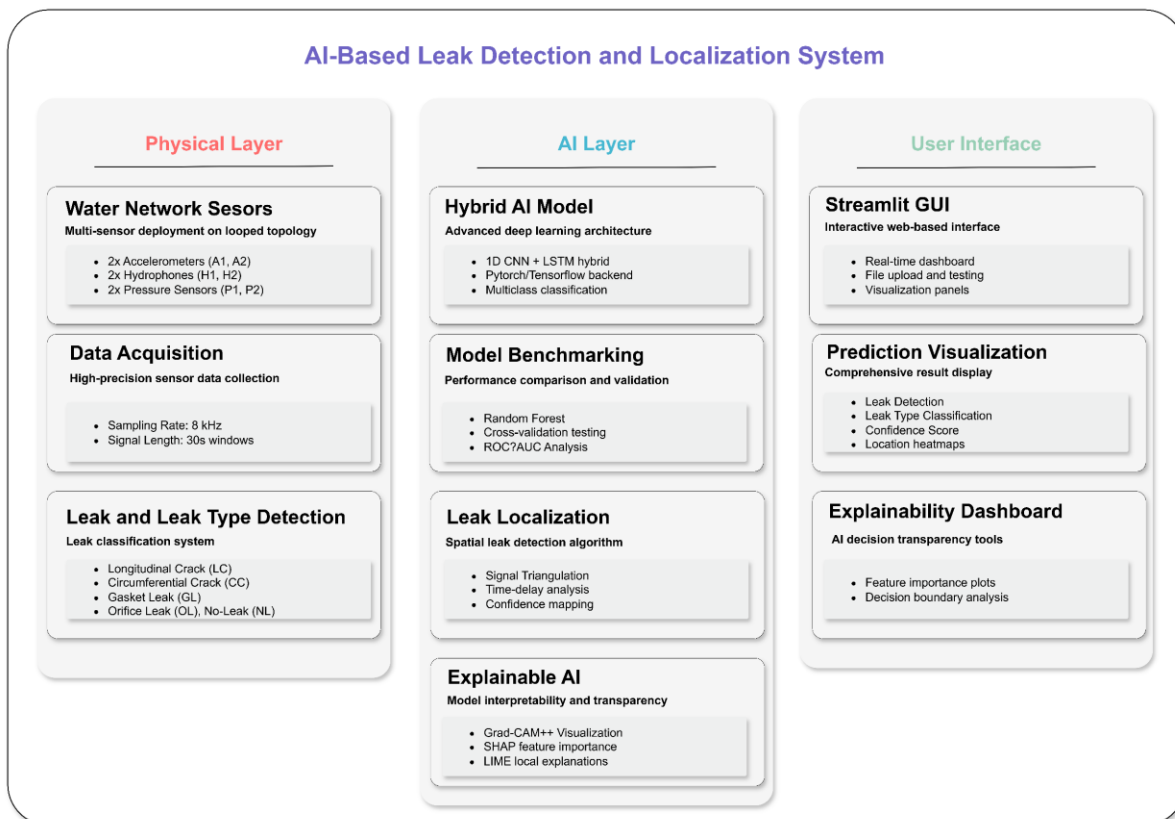
- **Simulation:** Will be Python-based; WNRT (Water Network Research Tool) is being evaluated for physics-based digital twin capability.
- The simulation will mimic real network conditions & sensor placements. Allow injections of synthetic leaks at various locations. Generate corresponding sensor signals for model testing.

Current Achievements

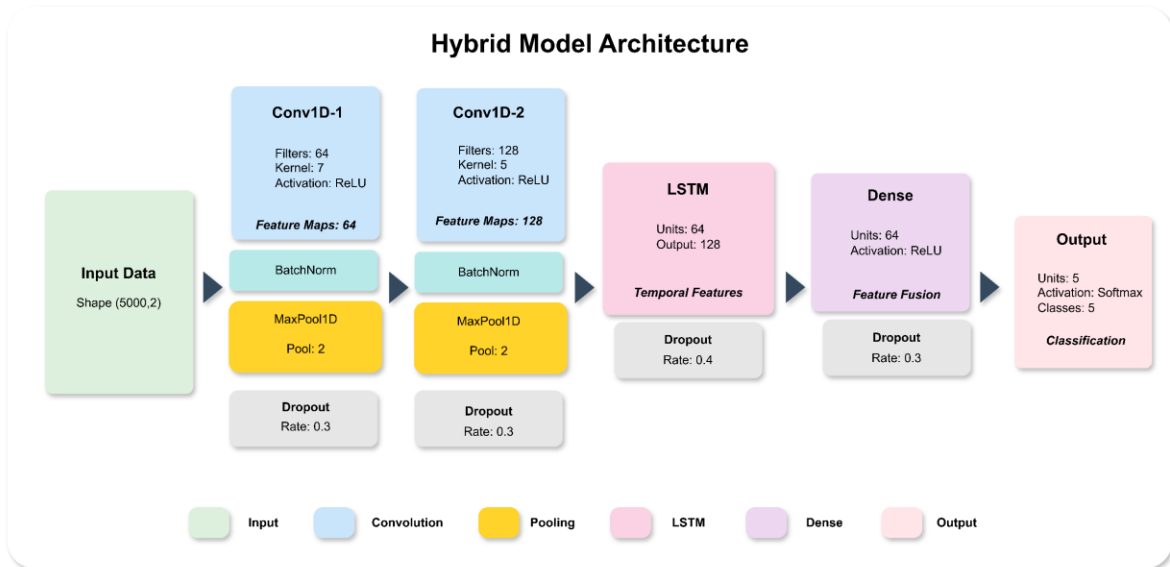
1. Preprocessing pipeline completed (scaling, CWT, log normalization).
2. Baseline hybrid CNN–LSTM model implemented & tested.
3. Integration of attention mechanism explored.
4. Achieved 88% accuracy (Pressure + Accelerometer) in baseline model.
5. Transfer Learning experiment reached 84% accuracy in just 5 epochs.
6. Prepared conceptual diagrams for architecture & training flow. *(Insert Figures Here)*
7. Defined a clear plan for simulation/digital testbed & GUI.
8. Performance metrics & specifications agreed with client.

Designed the following figures:

(a) Conceptual diagram



(b) Hybrid model



c) Training flow

